

Tan Yong Jie

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Robotics Systems Engineering Student (SIT, Year 2) seeking internship in robotics software or autonomous systems. Built and delivered proof-of-concept systems to industry stakeholders across autonomous navigation, computer vision, and embedded systems. Two-time WorldSkills Singapore Medallion of Excellence recipient.

PROJECTS

Omnidirectional Perception System (OPS) - Technical Lead (Software) Jan - Mar 2026

ROS2 Humble, Python, OpenCV, Tesseract OCR, pyzbar, SQLite, Raspberry Pi | [GitHub](#)

- Delivered PoC omnidirectional barcode scanning system to industry stakeholder, addressing manual scanning errors across inbound and sorting operations.
- Designed SmartMatcher engine fusing OCR and barcode data per item, handling degraded and misoriented barcodes without failure.
- Optimised Pi camera pipeline to meet sub-2.5s scan efficiency across all 3 cameras.

Autonomous Maze Escape Robot Sep - Dec 2025

ROS2 Humble, Python, SLAM Toolbox, Nav2, Gazebo, Turtlebot3 | [GitHub](#)

- Built autonomous maze escape system navigating TurtleBot3 from random spawn to exit. Initially targets a fixed far goal, then switches to max LiDAR range as goal when forward path is clear.
- Developed LiDAR exit detection requiring 80%+ forward infinite readings over 5s window; zero false positives across all test runs.
- Built multi-directional recovery system analysing 4-direction clearance; achieved 95%+ stuck-recovery success rate.

Autonomous Indoor Navigation - Team Leader (Software) May - Aug 2025

ROS1 Noetic, Python, RTAB-Map, Navigation Stack, RViz, Jetson Nano, AgileX LIMO | [GitHub](#)

- Led 6-person team delivering autonomous navigation for AgileX LIMO across all 8 randomised plots in a shared 4,500×3,990mm arena.
- Implemented custom recovery behavior (costmap clearing, in-place rotation, reverse escape) using MoveBaseAction and /move_base/make_plan.
- Fused depth camera, 2D LiDAR, IMU, and wheel encoders for AMCL localisation and real-time SLAM; no drift-induced failures across full arena.

EXPERIENCE

Singapore Institute of Technology - Student Technologist Aug 2025 to Apr 2026

- Migrated 16 AgileX LIMO robots from Jetson Nano to Orin Nano, to support ROS2 Humble deployment across university teaching labs.
- Validated full stack compatibility (chassis driver, LiDAR, depth camera); zero regressions post-migration.

COMPETITION

Worldskills Singapore - Mechatronics Medallion of Excellence (2020 & 2021)

- Awarded Medallion of Excellence twice at national level (2020 & 2021) in WorldSkills Singapore Mechatronics, competing in 3 national competitions (2018, 2020, 2021).
- Designed, assembled, and programmed modular industrial automation systems in a 2-person team, covering PLC, HMI, pneumatics, and robotics using Ladder Diagram and SFC.

CERTIFICATIONS

Certified SOLIDWORKS Associate (CSWA) — Dassault Systèmes

EDUCATION

B.Eng (Hons) Robotics Systems Engineering (Singapore Institute of Technology) Sep 2024 - Present (Expected 2028)

Diploma in Robotics & Mechatronics Engineering (Nanyang Polytechnic) Apr 2019 - Mar 2022

Director's List (3 semesters) · WorldSkills Singapore Mechatronics 2020 & 2021

LANGUAGES & TECHNOLOGIES

Robotics & Navigation: ROS1, ROS2, Nav2, SLAM Toolbox, RTAB-Map, TurtleBot3, AgileX LIMO

Computer Vision: OpenCV, Tesseract OCR, pyzbar, LiDAR, Depth Camera

Programming: Python, C, C++, Linux, Git

Embedded & Industrial: STM32, Raspberry Pi, Jetson Nano, TwinCAT PLC, Panasonic PLC/HMI (LD, SFC)

Tools: RViz, Gazebo, SQLite

Languages: English (fluent), Mandarin (spoken)